

Revised Load Interconnection System Impact Study Report

For Queue Project Q15-0476

Medway Hydro 100 kV
March 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0476. The Interconnection Customer proposes a 224.1 MW large load (data center) facility interconnecting at the Medway Hydro 100 kV substation in the Michigan area. The request was received on May 11, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 208 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,187,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: El Sol Substation – W Loop Tap Station 100.0 kV
Contingency: Imperial Vly – Medway Hydro Substation 100.0 kV
Loading: 124.0% (Rating: 135 MVA)
DFAX: 100.0%
Pre-project loading: 41.5%
2. Facility: Fossil – Aep E Watervliet Substation 100.0 kV line
Contingency: Imperial Vly – Medway Hydro 100.0 kV L
Loading: 118.9% (Rating: 125 MVA)
DFAX: 57.3%
Pre-project loading: 16.0%
3. Facility: Imperial Vly – Medway Hydro 100.0 kV
Contingency: Sister Lakes – El Sol 100.0 kV
Loading: 116.7% (Rating: 159 MVA)
DFAX: 100.0%
Pre-project loading: 24.2%
4. Facility: El Sol – Sister Lakes Substation 100.0 kV L
Contingency: Imperial Valley – Medway Hydro 100.0 kV L
Loading: 114.1% (Rating: 163 MVA)
DFAX: 100.0%
Pre-project loading: 23.7%
5. Facility: Aep E Watervliet – Medway Hydro Substation 100.0 kV line
Contingency: Imperial Vly – Medway Hydro 100.0 kV
Loading: 113.9% (Rating: 123 MVA)
DFAX: 57.3%
Pre-project loading: 9.2%

6. Facility: C.H. Dexter – Santa Fe Energy 100.0 kV L
 Contingency: Flood Street – Cook 500.0 kV
 Loading: 105.0% (Rating: 150 MVA)
 DFAX: 16.2%
 Pre-project loading: 80.9%
7. Facility: Ingledow – Fossil Substation 100.0 kV
 Contingency: Fossil – Aep East Waterviet 100.0 kV
 Loading: 102.9% (Rating: 118 MVA)
 DFAX: 15.6%
 Pre-project loading: 73.3%
8. Facility: Santa Fe Energy – Barrigada 100.0 kV
 Contingency: Flood St – Cook 500.0 kV
 Loading: 102.3% (Rating: 185 MVA)
 DFAX: 16.2%
 Pre-project loading: 82.7%
9. Facility: Sister Lakes – Berrien Springs 100.0 kV Line
 Contingency: Imperial Vly – Medway Hydro Substation 100.0 kV
 Loading: 100.7% (Rating: 153 MVA)
 DFAX: 64.7%
 Pre-project loading: 5.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
El Sol Substation – W Loop Tap Station 100.0 kV	Rebuild line with higher-capacity conductor	\$3,954,000	54% (\$2,125,000)
Fossil – Aep E Watervliet Substation 100.0 kV line	Rebuild line with higher-capacity conductor	\$12,089,000	100% (\$12,089,000)
Imperial Vly – Medway Hydro 100.0 kV	Rebuild line with higher-capacity conductor	\$10,440,000	54% (\$5,612,000)
El Sol – Sister Lakes Substation 100.0 kV L	Rebuild line with higher-capacity conductor	\$3,026,000	54% (\$1,626,000)
Aep E Watervliet – Medway Hydro Substation 100.0 kV line	Rebuild line with higher-capacity conductor	\$4,933,000	100% (\$4,933,000)
C.H. Dexter – Santa Fe Energy 100.0 kV line	Reconductor line	\$4,059,000	57% (\$2,298,000)
Ingledow – Fossil 100.0 kV Line	Reconductor line	\$1,772,000	23% (\$402,000)
Barrigada – Santa Fe Energy 100.0 kV	Reconductor line	\$886,000	57% (\$502,000)
Berrien Springs – Sister Lakes 100.0 kV	Reconductor line	\$2,977,000	54% (\$1,600,000)
	Total allocated to project		\$31,187,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.